Practice Set 3: Wrong-Answer Notes

CFA Level II · 6 wrong answers, one per page

46 / 52

Practice set 3 (88.5%)

Q49-52 were marked against the version you sat (the credit vignette before it was re-skinned on 11 Nov).

Score by topic

Topic	Score
Portfolio Management	5/7
Alternative Investments	3/4
Ethical and Professional Standards	3/4
Quantitative Methods	5/6
Fixed Income	14/15
Derivatives	4/4
Financial Statement Analysis	2/2
Economics	1/1
Corporate Issuers	9/9

Questions at a glance

Q	Topic	You	Ans
Q9	Ethics	C	B
Q21	Alt	C	A
Q34	PM	B	A
Q39	Quant	C	B
Q43	PM	C	A
Q50	FI	A	B

Set 3 Q9 **Standard V(B): separate fact from opinion, and say how uncertain a forecast is**

You **C** · Answer **B**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Ethics: Standards of Professional Conduct

Question 9. Scott works for a regional brokerage firm. He estimates that Walkton Industries will increase its dividend by US$1.50 a share during the next year. He realizes that this increase is contingent on pending legislation that would, if enacted, give Walkton a substantial tax break. The US representative for Walkton's home district has told Scott that, although she is lobbying hard for the bill and prospects for its passage are favorable, concern of the US Congress over the federal deficit could cause the tax bill to be voted down. Walkton Industries has not made any statements about a change in dividend policy. Scott writes in his research report, "We expect Walkton's stock price to rise by at least US$8.00 a share by the end of the year because the dividend will increase by US$1.50 a share. Investors buying the stock at the current time should expect to realize a total return of at least 15% on the stock." According to the Standards:

OPTIONS

A	Scott violated the Standards because he used material inside information.		Not inside information: the facts came from a public representative and the company said nothing, so there is no material nonpublic information.
B	Scott violated the Standards because he failed to separate opinion from fact.	✓ Correct	Correct: V(B). The report states the dividend rise and an $8 price gain as certain, although both depend on the bill passing.
C	Scott violated the Standards by basing his research on uncertain predictions of future government action.	✗ Your pick	Your pick: basing research on an uncertain prediction of government action is allowed. The violation is how it was communicated.

Trigger "contingent on pending legislation"
"could cause the tax bill to be voted down"
"because the dividend will increase by US$1.50"

Rule next time: A forecast that depends on an uncertain event must not be written as if it were certain. V(B): separate fact from opinion and disclose the uncertainty. Relying on a prediction is fine; presenting it without the conditions is the violation.

FORMULA / RULE

> V(B): distinguish fact from opinion; include the important factors and limits of the analysis
> Words that signal a V(B) problem: "will", "at least", "guaranteed" attached to an uncertain event

WHY THE ANSWER IS B

1. The dividend increase is conditional on legislation that may fail.
2. The report says the price WILL rise "because the dividend will increase": opinion presented as fact.
3. **B**.

Apply to other questions: Ask: does the report show the condition behind the forecast? If not, V(B). Use of public, uncertain information is not itself a violation.

Set 3 Q21 NAVPS: capitalise NOI, then add other assets and subtract debt

You **C** · Answer **A**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Castell values Alpenhaus REIT. Next year's net operating income is CHF 32 million and the market capitalisation rate for similar properties is 6.4%. Alpenhaus also holds CHF 18 million of cash and other tangible assets, and has CHF 290 million of debt. It has 10 million shares.

Castell notes that Alpenhaus is highly leveraged and trades at a large discount to NAVPS, and that its debt matures next year.

Question 21. Alpenhaus's NAVPS is closest to:

OPTIONS

A	CHF 22.80	✓ Correct	Correct: (500 + 18 − 290) ÷ 10 = CHF 22.80.
B	CHF 51.80		CHF 51.80 = (500 + 18) ÷ 10 adds the cash but forgets to subtract the debt.
C	CHF 50.00	✗ Your pick	Your pick: CHF 50.00 = 500 ÷ 10 is the property value per share. Cash and debt were left out.

Trigger “market capitalisation rate for similar properties is 6.4%”
“CHF 18 million of cash and other tangible assets”
“has CHF 290 million of debt”

Rule next time: NAV is the equity value: property value (NOI ÷ cap rate) + cash and other assets − debt. Only then divide by the shares.

FORMULA / RULE

> Property value = NOI ÷ market cap rate (direct capitalisation, not depreciated cost)
> NAV = property value + other assets − liabilities; NAVPS = NAV ÷ shares

WHY THE ANSWER IS A

1. Property value = 32 ÷ 0.064 = 500.0.
2. NAV = 500 + 18 − 290 = 228.0.
3. NAVPS = 228 ÷ 10 = CHF 22.80: **A**.

Apply to other questions: Every item in the vignette (cash, debt) is a step. Value ≠ NAV until the balance sheet items are applied.

Set 3 Q34 Real rates follow REAL growth and its volatility, not nominal growth

You **B** · Answer **A**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Point 3: Country 1 has nominal GDP growth of 8.5% and inflation of 3.0%, with low volatility of growth. Country 2 has nominal GDP growth of 6.0% and inflation of 0.5%, with high volatility of growth.

Question 34. Based on Point 3, real interest rates are most likely higher in:

OPTIONS

A	Country 2, because real growth is the same but growth is more volatile	✓ Correct	Correct: real growth is 5.5% in both, and Country 2's growth is more volatile, so its real rate is higher.
B	Country 1, because nominal growth is higher	✗ Your pick	Your pick: Country 1 has higher NOMINAL growth, but nominal growth includes inflation and says nothing about the real rate.
C	neither country, because real growth is the same		Equal real growth does not mean equal real rates: volatility of growth also matters.

Trigger “nominal GDP growth of 8.5% and inflation of 3.0%”
“nominal GDP growth of 6.0% and inflation of 0.5%”
“low volatility” vs “high volatility of growth”

Rule next time: Convert to real growth first: nominal − inflation. If real growth is equal, the country with more volatile growth has the higher real interest rate.

FORMULA / RULE

Real growth ≈ nominal growth − inflation

Real rate ↑ with higher expected real growth and with higher volatility of real growth

ITRS ↓ (better off in the future) → real rates ↑

WHY THE ANSWER IS A

1. Country 1: 8.5 − 3.0 = 5.5%. Country 2: 6.0 − 0.5 = 5.5%.
2. Equal, so compare volatility: Country 2 is higher.
3. Real rates higher in Country 2: **A**.

Apply to other questions: Subtract inflation before comparing growth rates, then look at volatility.

Set 3 Q39 **AIC is for forecasting, BIC for goodness of fit; for both the LOWER value is better**

You **C** · Answer **B**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Exhibit 1: ANOVA for the Three-Factor Model

Source	Degrees of freedom	Sum of squares
Regression	3	0.0770
Error	56	0.0480
Total	59	0.1250

Greco also compares two candidate models. Model A has AIC −310 and BIC −298. Model B has AIC −314 and BIC −296.

Question 39. Based on AIC and BIC, which model is preferred for forecasting and which for goodness of fit?

OPTIONS

A	Model A for forecasting; Model B for goodness of fit		Reversed: A for forecasting and B for goodness of fit mixes up which criterion is for which purpose.
B	Model B for forecasting; Model A for goodness of fit	✓ Correct	Correct: AIC picks Model B (−314 < −310) for forecasting; BIC picks Model A (−298 < −296) for goodness of fit.
C	Model B for both	✗ Your pick	Your pick: Model B wins on AIC only. On BIC, Model A is lower.

Trigger “Model A has AIC −310 and BIC −298. Model B has AIC −314 and BIC −296”

Rule next time: AIC → forecasting (prediction), BIC → goodness of fit. Lower is better for both, and −314 is lower than −310.

FORMULA / RULE

AIC (lighter penalty for extra variables): better for forecasting
BIC: heavier penalty for extra variables: better for goodness of fit
Choose the model with the LOWEST value (with negative numbers, the more negative one)

WHY THE ANSWER IS B

1. AIC: −314 (B) < −310 (A) → B is better for forecasting.
2. BIC: −298 (A) < −296 (B) → A is better for goodness of fit.
3. **B**.

Apply to other questions: Compare each criterion separately; do not pick one model for both just because it wins on the first.

Set 3 Q43 Discount rate for rents: tenant credit risk is in the discount rate; faster rent growth is in the cash flow

You **C** · Answer **A**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Lucia Ferrante makes three further observations.

Observation 3: An investor compares a building leased to a government agency with a similar building leased to commercial tenants.

Question 43. For Observation 3, the commercial building's rental income should most likely be discounted at:

OPTIONS

A	a higher rate, because it includes a credit risk premium for the tenants	✓ Correct	Correct: higher rate, because the commercial tenants add a credit risk premium.
B	the same rate, because both include inflation and inflation uncertainty premiums		Both include inflation and inflation uncertainty, but the commercial lease also has the credit premium, so the rates are not the same.
C	a lower rate, because commercial rents rise faster	✗ Your pick	Your pick: rents rising faster would raise the cash flows (the numerator), not lower the discount rate.

Trigger “a building leased to a government agency”
“a similar building leased to commercial tenants”

Rule next time: Same building, different tenant credit quality: the commercial lease carries an extra credit risk premium φ, so its rents are discounted at a HIGHER rate.

FORMULA / RULE

Discount rate = Rf + π + θ + φ (inflation, inflation uncertainty, other risks such as default)
Value = net rental income ÷ (1 + r)^t

WHY THE ANSWER IS A

1. Government tenant: very low default risk, small φ.
2. Commercial tenants: higher default risk, larger φ → higher r.
3. **A**.

Apply to other questions: Tenant credit risk → discount rate (φ). Growth in rents → cash flows.

Set 3 Q50 **Credit spread from CVA: LGD = 1 − recovery rate**

You **A** · Answer **B**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Marangoni next asks Brandt to analyze a three-year bond, issued by Norvalle S.A., using an arbitrage-free framework. The bond's coupon rate is 4%, with interest paid annually and a par value of 100. In the analysis, Brandt makes the following three assumptions:

The annual interest rate volatility is 15%.

The recovery rate is one-quarter of the exposure each period.

The annual probability of default each year is 1.50%.

Exhibit 2

Par Curve Rates for Annual Payment Benchmark Government Bonds

Maturity	Coupon rate	Price	Discount factor	Spot rate	Forward rate
1	2.50%	100	0.975610	2.5000%	2.5000%
2	3.30%	100	0.936888	3.3133%	4.1331%
3	4.00%	100	0.887981	4.0396%	5.5076%

Exhibit 3

One-Year Binomial Interest Rate Tree for 15% Volatility (risk-neutral probability of 50% on each branch)

Date	0	1 (top)	1 (bottom)	2 (top)	2 (middle)	2 (bottom)
Rate	2.5000%	4.7526%	3.5208%	7.2932%	5.4029%	4.0026%

Question 50. Based on Brandt's assumptions and Exhibits 2 and 3, the credit spread on the Norvalle bond is closest to:

OPTIONS

A	0.7796%.	✗ Your pick	Your pick: 0.7796% is what you get with a loss given default of 50%, not 75%.
B	1.1782%.	✓ Correct	Correct: LGD 75% gives a spread of 1.1782%.
C	1.5828%.		1.5828% assumes no recovery (LGD 100%).

Trigger “The recovery rate is one-quarter of the exposure each period.”
“The annual probability of default each year is 1.50%.”
“the relevant binomial interest rate tree is presented in Exhibit 3”: value the bond on the tree (VND)

Rule next time: Recovery of one-quarter means LGD = 1 − 1/4 = 75%. Then CVA = Σ DF × PD × LGD × expected exposure, fair value = VND − CVA, spread = YTM(fair value) − benchmark YTM.

FORMULA / RULE

LGD = 1 − recovery rate
CVA = Σ DF(benchmark) × PD(t) × LGD × expected exposure(t)
PD(t) = h × (1 − h)^(t−1) for a constant hazard rate h
Spread = YTM of (VND − CVA) − benchmark YTM

WHY THE ANSWER IS B

1. VND from the tree = 100.0000 (coupon equals the 3-year par rate).
2. CVA = 1.1250 + 1.0648 + 1.0080 = 3.1978.
3. Fair value = 96.8022, YTM = 5.1782%.
4. Spread = 5.1782% − 4.0000% = 1.1782%: **B**.

Apply to other questions: Recovery rate given → convert to LGD before anything else; read “one-quarter of the exposure” as recovery, not loss.